Experiment – 03

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Branch: BE-CSE(AIML)

Subject: Advanced Database Management System

Question 1:

You are given the Employee schema, where each employee belongs to a department and has a performance score. The task is to find the highest score for each respective department, and update the final score for every employee in that department to be that department's highest score.

Employee Table (Input):

EID	DEPT	SCORES
1	D1	1
2	D1	5.28
3	D1	4
4	D2	8
5	D1	2.5
6	D2	7
7	D3	9
8	D4	10.2

Your text here 1

Example Output:

EID	DEPT	SCORES
1	D1	5.28
2	D1	5.28
3	D1	5.28
4	D2	8
5	D1	5.28
6	D2	8
7	D3	9
8	D4	10.2

Solution:

```
SELECT
  EID,
  DEPT,
  MAX(SCORES) OVER (PARTITION BY DEPT) AS SCORES
FROM EMPLOYEE
ORDER BY EID;
```

Question 2:

Sales Analysis — Part 1: A company maintains records of products sold by different salespersons. The management wants to prepare a sales performance report by identifying the salesperson(s) who generated the highest overall sales revenue. If multiple salespersons have the same highest total sales amount, all of them must be included in the final report. Return the seller ID(s) satisfying this condition.

SalesPerson Table:

seller_id	name	city	commission
1	John	New York	15
2	Amy	Los Angeles	13
3	Mark	Chicago	12
4	Pam	Boston	15

Company Table:

com_id	name	city
1	RED	Boston
2	ORANGE	New York
3	YELLOW	Boston
4	GREEN	Austin

Orders Table:

order_id	seller_id	amount
1	1	1200
2	1	800
3	2	2500
4	3	1500
5	2	700
6	3	2000
7	4	3000
8	4	200

Example Output:

seller_id
3

Solution:

```
WITH SELLER_TOTALS AS (
  SELECT
    SELLER_ID,
    SUM(AMOUNT) AS TOTAL_SALES
  FROM ORDERS
  GROUP BY SELLER_ID
)
SELECT
  SELLER_ID
```

```
FROM SELLER_TOTALS
WHERE TOTAL_SALES = (SELECT MAX(TOTAL_SALES) FROM SELLER_TOTALS);
```

Question 3:

A company maintains employee salary records for different departments. The management wants to prepare a report containing employees whose salaries fall among the top three highest distinct salary values within their respective departments. If multiple employees receive the same qualifying salary, all such employees must be included in the final report. Display the department name, employee name, and salary for every qualifying employee.

Employee Table:

ID	Name	Salary	DepartmentId
1	Joe	85000	1
2	Henry	80000	2
3	Sam	60000	2
4	Max	90000	1
5	Janet	69000	1
6	Randy	85000	1
7	Will	70000	1

Department Table:

ID	DEPT_NAME
1	IT
2	Sales

Example Output:

Department	Employee	Salary
IT	Max	90000
IT	Joe	85000
IT	Randy	85000
IT	Will	70000
Sales	Henry	80000
Sales	Sam	60000

Solution:

```
WITH RANKED_SALARIES AS (
  SELECT
    D.DEPT_NAME,
    E.NAME,
    E.SALARY,
    DENSE_RANK() OVER (
      PARTITION BY E.DEPARTMENTID ORDER BY E.SALARY DESC
    ) AS SALARY_RANK
```

```
  FROM EMPLOYEE AS E
  JOIN DEPARTMENT AS D
    ON E.DEPARTMENTID = D.ID
)
SELECT
  DEPT_NAME AS "Department",
  NAME AS "Employee",
  SALARY
FROM RANKED_SALARIES
WHERE SALARY_RANK <= 3
ORDER BY DEPT_NAME;
```